

DATA CLEANING & REPORTING AUTOMATION

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Introduction

In many real-world datasets, the data is not always clean and organized. There can be missing values, repeated records, formatting issues, and inconsistent information. These problems can affect the accuracy of analysis and reporting.

The main purpose of this project is to automate the process of cleaning data and generating reports using Python and Power BI. By automating these tasks, the work becomes faster, easier, and more efficient.

Objective of the Project

The objective of this project is to:

- Clean and preprocess raw data
- Handle missing values and duplicate records
- Standardize inconsistent text data
- Generate automated reports and charts
- Create a visual dashboard for analysis

Dataset Used

For this project, a salary dataset was used. The dataset contains employee-related information such as:

- Age
- Gender
- Education Level
- Job Title
- Years of Experience
- Salary

While checking the dataset, some missing values and duplicate records were found, which needed to be cleaned before analysis.

Technologies Used

Python

Python was used to automate the entire data cleaning process.

Pandas

The Pandas library helped in:

- Reading the dataset
- Handling missing values
- Removing duplicates
- Cleaning text formatting
- Creating summary reports

Matplotlib

Matplotlib was used to generate charts and visual summaries automatically.

Power BI

Power BI was used to create an interactive dashboard for data visualization and reporting.

Process Followed

Loading the Dataset

The dataset was imported into Python using Pandas.

Handling Missing Values

Missing numerical values were replaced using average values, and missing text values were filled with “Unknown”.

Removing Duplicate Records

Duplicate rows were identified and removed from the dataset to improve data quality.

Cleaning Text Data

Text formatting issues such as extra spaces and inconsistent capitalization were corrected.

Generating Reports

A summary report was generated automatically using Python.

Creating Visualizations

Charts for Age, Salary, and Years of Experience were created automatically and saved.

Dashboard Creation

A Power BI dashboard was developed to display employee salary analysis and insights visually.

Outcome of the Project

This project helped in understanding how data cleaning and reporting automation work in real-world scenarios. The workflow reduced manual effort and improved the quality of the dataset. Automated reports and dashboards also made the analysis easier to understand.

Conclusion

Through this project, I learned how to clean raw datasets, automate preprocessing tasks, and create reports using Python and Power BI. This project also improved my understanding of data analysis workflows and reporting techniques.